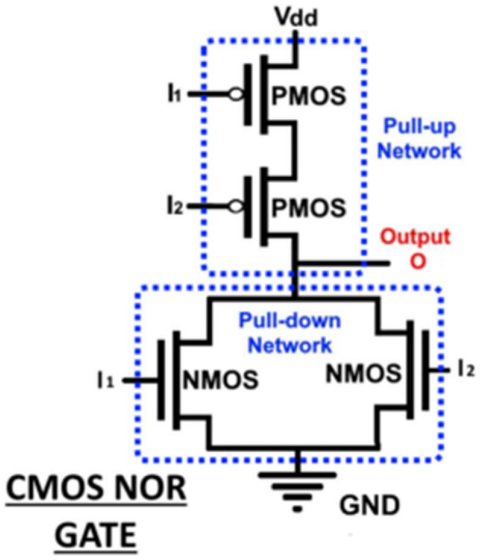
Lecture 32: CMOS NAND and NOR Gate

CMOS NOR Logic

CMOS (Complementary metal oxide semiconductor) is mostly used in designing a variety of logic circuit as they are most economical,

The design of NOR gate using CMOS is given below:



In this design, NMOS & PMOS are both used together.

PMOS part is known as Pull-Up Network (PUN) & NMOS part is known as Pull-Down Network (PDN).

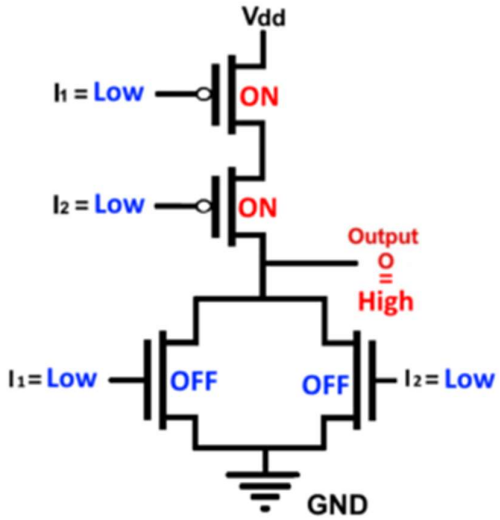
In the Pull-up network part, two PMOS are connected in series. The source of one NMOS is connected with the drain of the second PMOS & the Drain of first PMOS is directly connected to the Vdd.

In Pull-down network part, two NMOS are connected in parallel with each other whose source is connected to the GND.

These both networks are connected together as shown in the figure above. The gate of one PMOS and NMOS is connected together to make one input & the output is taken between these two networks (PUN & PDN).

Case 1

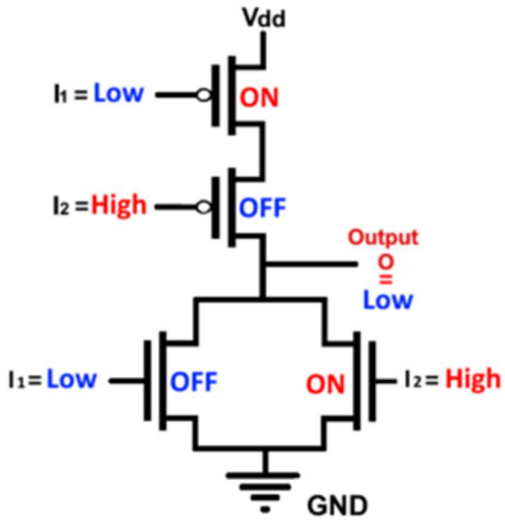
When both of the input is logic Low i.e. I_1 & I_2 = 0v.



The PMOS will switch ON & NMOS will switch OFF. Only the pull-up network (PUN) activates & As a result, it creates a conductive path from the Vdd to the Output. Due to which, the output becomes logic High.

Case 2

When one of the input is logic high and the other is logic Low i.e. I_1 = 0v & I_2 = 5v.

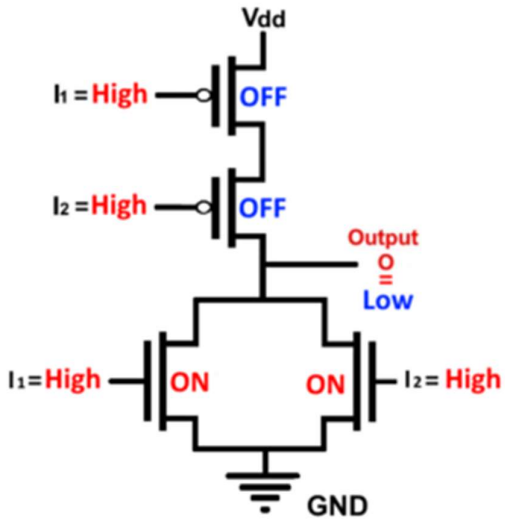


In this case, one PMOS & one NMOS will switch ON in both PUN & PDN network. It will result in activating only the pull-down network (PDN) because of the activated NMOS in

parallel. This configuration results in a direct path from GND to output. Due to which the output becomes logic Low.

Case 3

When both of the input is logic High i.e. I_1 & I_2 = 5v

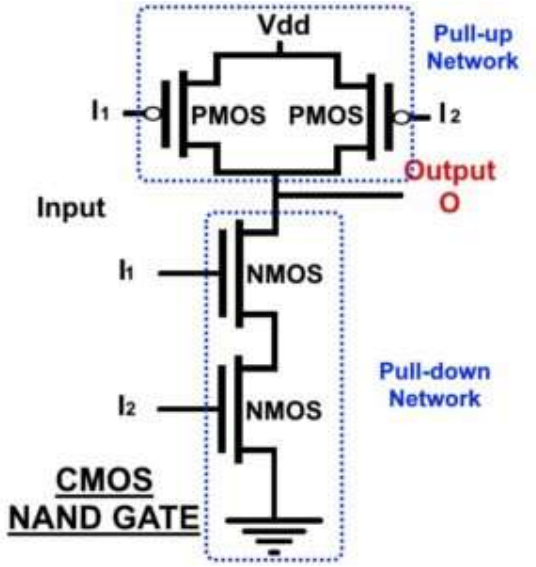


The PMOS will switch OFF & the NMOS will switch ON because their gate input is logic High. As a result, only the Pull-down network (PDN) activates which creates a direct path from GND to output. Thus the output becomes logic Low.

CMOS NAND Logic

CMOS (Complementary metal oxide semiconductor) is mostly used in designing a variety of logic circuit as they are most economical,

The design of NAND gate using CMOS is given below:



In this design, NMOS & PMOS are both used together.

PMOS part is known as Pull-Up Network (PUN) & NMOS part is known as Pull-Down Network (PDN).

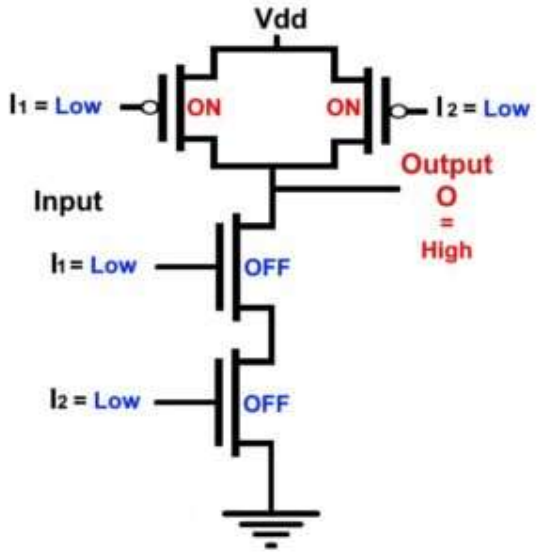
In Pull-up network part, two PMOS are connected in parallel with each other whose drain is connected to supply voltage Vdd.

In Pull-down network part, two NMOS are connected in series. The source of one NMOS is connected with the drain of the second NMOS & the Source of second NMOS is directly connected to the GND.

These both networks are connected together as shown in the figure above. The gate of one PMOS and NMOS is connected together to make one input & the output is taken between these two networks (PUN & PDN).

Case 1

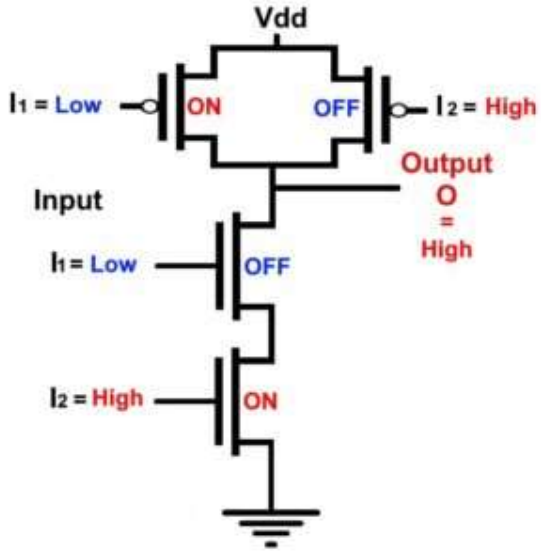
When both of the input is logic Low i.e. I_1 & I_2 = 0v.



The PMOS will switch ON & NMOS will switch OFF. Only the pull-up network (PUN) activates & the pull-dewn network (PDN) breaks the path between Vdd & GND. As a result, it creates a conductive path from the Vdd to the Output. Due to which, the output becomes logic High.

Case 2

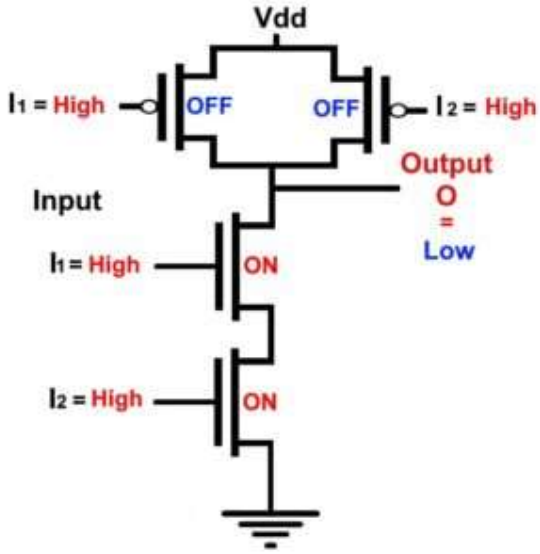
When one of the input is logic high and the other is logic Low i.e. I_1 = 0v & I_2 = 5v.



In this case, one PMOS & one NMOS will switch ON. It will result in activating only the pull-up network (PUN) which means a direct path from Vdd to output. As a result, the output becomes logic High.

Case 3

When both of the input is logic High i.e. I_1 & I_2 = 5v



The PMOS will switch OFF because the gate input is High & the NMOS will switch ON as their gate input is logic High. As a result, only Pull-down network (PDN) activates which creates a direct path from GND to output. Thus the output becomes logic Low.

----------END OF LECTURE----------

REFERENCES

1. **Text book**: **T1.** Modern Digital Electronics by R. P. Jain (Tata McGraw Hill)
2**. Reference book**: **R1.** Fundamental of digital circuits by A. Anand Kumar (PHI)

[redacted]
